LMA ASSIGNMENT 1

Baseline Retrieval vs. Re-Ranking

Surya Teja Thodupunuri
Roll Number: 2025701054

18TH OCTOBER 2026

GitHub Repository: https://github.com/suryatejathodupunuri/lma_a1

1. Architecture Comparison

The dataset is the 32-page *Spirit Island* core rulebook. It contains general rules, exceptions, and restrictions in different sections. The extracted text was divided into 144 page-based chunks, with a maximum of 220 tokens and an overlap of 40 tokens. Both pipelines use the same chunks.

The baseline uses sentence-transformers/all-MiniLM-L6-v2, a Bi-Encoder. It embeds the question and each chunk independently into 384-dimensional vectors. Chunk embeddings are stored in a NumPy index. The vectors are normalized, so their dot product equals cosine similarity. The baseline retrieves the top ten chunks and gives the first three to the LLM to produce Answer 1.

The second pipeline gives those same ten question–chunk pairs to cross-encoder/ms-marco-MiniLM-L6-v2. The Cross-Encoder reads the question and passage together. Joint self-attention lets their tokens interact before it assigns a relevance score. The candidates are sorted by descending score, and the top three are given to the LLM to produce Answer 2.

Both answers use Gemini 3.8 Flash with the same prompt template, temperature 0, a 2048-token output limit, and low thinking level. Only the three context chunks change. The reference answers are not sent to the LLM.

2. Failure Queries and Rank Shifts

Three questions were selected through exploratory testing. In each case, the critical clause was initially outside the top three but inside the top ten. Re-ranking brought it into the context used for Answer 2.

Case	Question	Critical evidence	Rank shift
F1	When I choose not to resolve a card, do I lose its Elements and get a refund?	`chunk_0084` p. 17	4 → 2
F2	Can I forget the Major Power I just gained instead of an old card?	`chunk_0090` p. 18	6 → 2
F3	If I increase a Slow Power to Fast, may I use it in both phases?	`chunk_0086` p. 17	5 → 1

The following comparisons show the actual saved LLM outputs. The baseline failures were incomplete answers or abstentions, rather than invented rules. These selected stress tests do not estimate overall accuracy.

2. Qualitative Answer Comparison (continued)

F1. Missing refund rule

Question: When I choose not to resolve a card, do I lose its Elements and get a refund?

chunk_0084 · PDF page 17 · Rank 4 → 2

Answer 1 — baseline (actual LLM output)

Based on the provided excerpts:

- **Elements:** No, you do not lose the Elements. You immediately gain all Elements from played Power Cards, and a Power Card can be played purely for its Elements even if you skip its text effect [chunk_0050]. Elements only go away during the "Time Passes" phase at the end of the turn [chunk_0058].
- **Refund:** The provided excerpts do not state whether or not you get an Energy refund when you choose not to resolve a card (only mentioning that you must pay Energy immediately when the card is played) [chunk_0050]. Therefore, the refund part of the question cannot be answered.

Answer 2 — re-ranked (actual LLM output)

No. If you choose to skip using a Power Card's effect, you do not get your Energy back (no refund), but you do still get to use the Elements granted by the card [chunk_0084, chunk_0085].

What changed: The baseline first retrieved an end-of-turn rule about Elements disappearing. It also found that effects could be skipped, but missed the explicit refund rule. Answer 1 therefore covered the Elements but left the refund unanswered. Re-ranking added the missing clause, so Answer 2 correctly covered both parts.

F2. General rule without the exception

Question: Can I forget the Major Power I just gained instead of an old card?

chunk_0090 · PDF page 18 · Rank 6 → 2

Answer 1 — baseline (actual LLM output)

Based on the provided excerpts, the rules state that after you get a new Major Power, you "must Forget - permanently lose - one of your Power Cards" [chunk_0087, chunk_0133]. However, the excerpts do not explicitly state whether the newly gained Major Power qualifies as "one of your Power Cards" that can be chosen to be forgotten. Therefore, it cannot be definitively answered from the text whether you can forget the Major Power you just gained instead of an older card.

Answer 2 — re-ranked (actual LLM output)

Yes, you can forget any of your cards, including the Major Power you just chose [chunk_0090].

What changed: The baseline found the general requirement to Forget a card after gaining a Major Power. It missed the clarification that the newly chosen card itself can be Forgotten. Answer 1 did not commit to an answer. Once the clarification entered the top three, Answer 2 correctly answered yes.

F3. Missing phase restriction

Question: If I increase a Slow Power to Fast, may I use it in both phases?

chunk_0086 · PDF page 17 · Rank 5 → 1

Answer 1 — baseline (actual LLM output)

Based on the provided excerpts, the text mentions that changes to a Power carry over if repeated (such as making a Power Fast) and that a Fast Power cannot be delayed to the Slow Phase ("Use it now or forego it") [chunk_0050, chunk_0093]. However, the excerpts do not explicitly state whether changing a Slow Power to Fast allows you to use it in both phases. Therefore, this specific question cannot be definitively answered from the provided text.

Answer 2 — re-ranked (actual LLM output)

No. If something makes a Slow Power Fast, you only get to use it during the Fast Phase, not both [chunk_0086].

What changed: The baseline found related passages about Fast Powers and repeating Powers. It missed the rule that a Slow Power made Fast is used only in the Fast Phase. Answer 1 said the available text was not enough. Re-ranking brought the restriction to first place, and Answer 2 correctly ruled out using both phases.

3. Latency Profiling

Retrieval and re-ranking ran on CPU with four PyTorch threads. Each question had one warm-up followed by five measured trials. Within each trial, both pipeline timings used the same initial retrieval. The advanced time adds re-ranking and top-three selection. Downloads, indexing, model loading, and LLM generation were excluded.

The table reports milliseconds. Baseline and total times show mean ± sample standard deviation; the re-ranking column shows its mean.

Case	Baseline retrieval	Re-ranking only (mean)	Retrieval + re-ranking
F1	4.91 ± 0.01	159.61	164.62 ± 1.43
F2	4.67 ± 0.03	140.10	144.86 ± 0.22
F3	5.14 ± 0.87	142.97	148.20 ± 3.03

The baseline took about 5 ms, compared with about 145–165 ms for retrieval plus re-ranking. The extra time allowed the system to include the missing rule clauses. For these three cases, that led to complete, supported answers.

4. Why Not Use a Cross-Encoder on the Entire Collection?

A Cross-Encoder scores a passage in relation to a particular question. Its relevance scores cannot be computed once and reused for every new question. Applying it to the whole collection would require a model evaluation for every question–chunk pair. That cost grows with both the collection size and the number of questions.

A Bi-Encoder lets the document vectors be computed ahead of time. For a new question, only its embedding needs to be computed before searching the stored vectors. The two-stage approach uses this faster search to narrow the collection, then applies the Cross-Encoder to only ten candidates. Here, that means ten pair evaluations instead of 144 per question. The limitation is that re-ranking cannot recover a relevant chunk that was missing from the initial top ten.